

R1 Forces and Newton's laws

Name: _____

Class: _____

Date: _____

Time: **341 minutes**

Marks: **286 marks**

Comments:

EXAM PAPERS PRACTICE

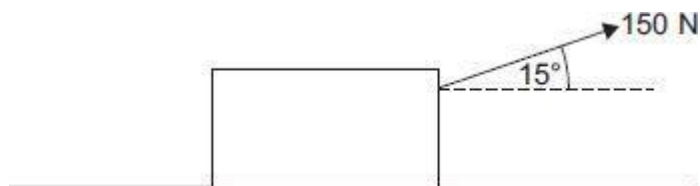
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Q1.

In this question use $g = 9.8 \text{ m s}^{-2}$

A boy attempts to move a wooden crate of mass 20 kg along horizontal ground. The coefficient of friction between the crate and the ground is 0.85

- (a) The boy applies a horizontal force of 150 N. Show that the crate remains stationary. (3)
- (b) Instead, the boy uses a handle to pull the crate forward. He exerts a force of 150 N, at an angle of 15° above the horizontal, as shown in the diagram.



Determine whether the crate remains stationary.

Fully justify your answer.

(5)
(Total 8 marks)

Q2.

In this question use $g = 10 \text{ m s}^{-2}$, giving your final answers to an appropriate degree of accuracy.

A man of mass 80 kg is travelling in a lift.
The lift is rising vertically.



The lift decelerates at a rate of 1.5 m s^{-2}

Find the magnitude of the force exerted on the man by the lift.

(Total 3 marks)

Q3.

A particle, of mass 400 grams, is initially at rest at the point O.

The particle starts to move in a straight line so that its velocity, $v \text{ m s}^{-1}$, at time t seconds is given by

$$v = 6t^2 - 12t^3 \text{ for } t > 0$$

- (a) Find an expression, in terms of t , for the force acting on the particle.

(3)

- (b) Find the time when the particle next passes through O.

(5)

(Total 8 marks)

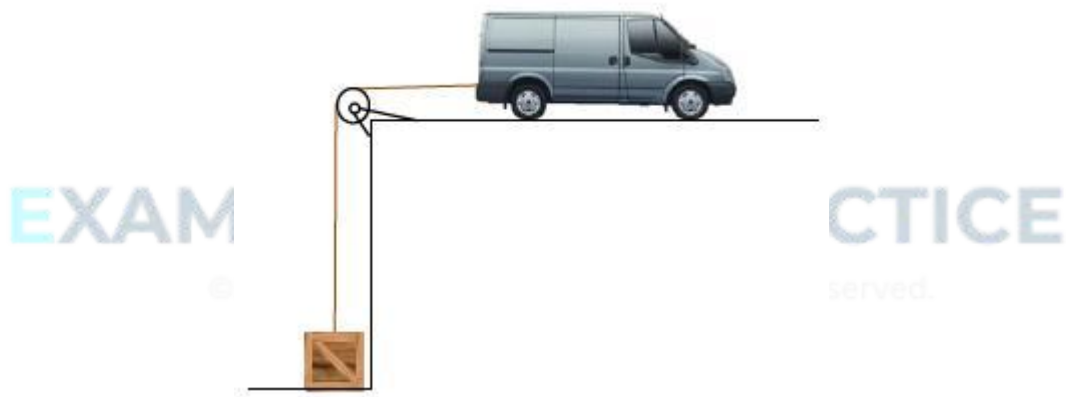
Q4.

In this question use $g = 9.8 \text{ m s}^{-2}$, giving your final answers to an appropriate degree of accuracy.

A van of mass 1300 kg and a crate of mass 300 kg are connected by a light inextensible rope.

The rope passes over a light smooth pulley, as shown in the diagram.

The rope between the pulley and the van is horizontal.



Initially, the van is at rest and the crate rests on the lower level. The rope is taut.

The van moves away from the pulley to lift the crate from the lower level.

The van's engine produces a constant driving force of 5000 N.

A constant resistance force of magnitude 780 N acts on the van.

Assume there is no resistance force acting on the crate.

- (a) (i) Draw a diagram to show the forces acting on the crate while it is being lifted.

(1)

- (ii) Draw a diagram to show the forces acting on the van while the crate is being lifted.

(1)

- (b) Show that the acceleration of the van is 0.80 m s^{-2} (4)
- (c) Find the tension in the rope. (2)
- (d) Suggest how the assumption of a constant resistance force could be refined to produce a better model. (1)
- (Total 9 marks)

Q5.

A uniform rod, AB , has length 3 metres and mass 24 kg.

A particle of mass M kg is attached to the rod at A .

The rod is balanced in equilibrium on a support at C , which is 0.8 metres from A .



Find the value of M .

(Total 2 marks)

Q6.

The three forces $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$ are acting on a particle.

$$\mathbf{F}_1 = (25\mathbf{i} + 12\mathbf{j}) \text{ N}$$

$$\mathbf{F}_2 = (-7\mathbf{i} + 5\mathbf{j}) \text{ N}$$

$$\mathbf{F}_3 = (15\mathbf{i} - 28\mathbf{j}) \text{ N}$$

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are horizontal and vertical respectively.

The resultant of these three forces is $\mathbf{F}$ newtons.

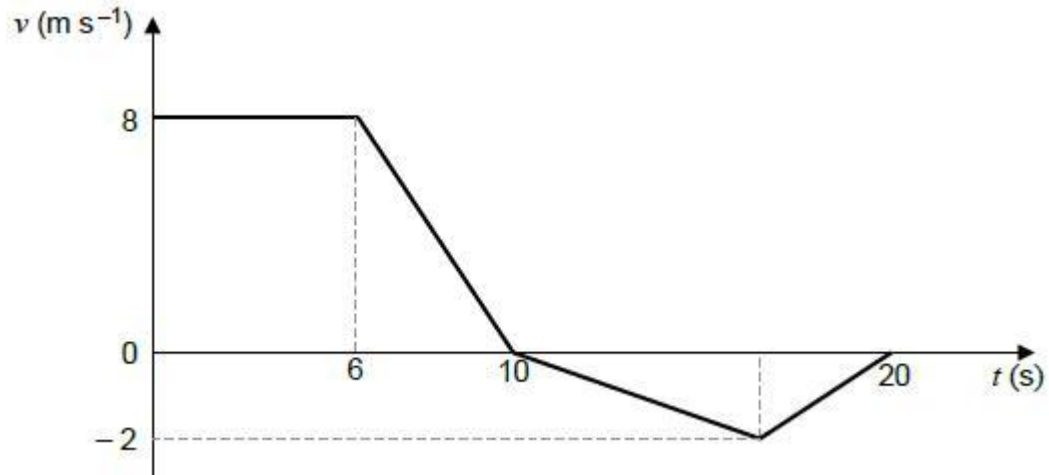
- (a) (i) Find the magnitude of $\mathbf{F}$, giving your answer to three significant figures. (2)
- (ii) Find the acute angle that $\mathbf{F}$ makes with the horizontal, giving your answer to the nearest 0.1° . (2)
- (b) The fourth force, $\mathbf{F}_4$, is applied to the particle so that the four forces are in equilibrium. Find $\mathbf{F}_4$, giving your answer in terms of $\mathbf{i}$ and $\mathbf{j}$. (1)
- (Total 5 marks)

Q7.

The graph below models the velocity of a small train as it moves on a straight track for 20 seconds.

The front of the train is at the point A when $t = 0$

The mass of the train is 800kg.



- (a) Find the total distance travelled in the 20 seconds. (3)
 - (b) Find the distance of the front of the train from the point A at the end of the 20 seconds. (1)
 - (c) Find the maximum magnitude of the resultant force acting on the train. (2)
 - (d) Explain why, in reality, the graph may not be an accurate model of the motion of the train. (1)
- (Total 7 marks)**

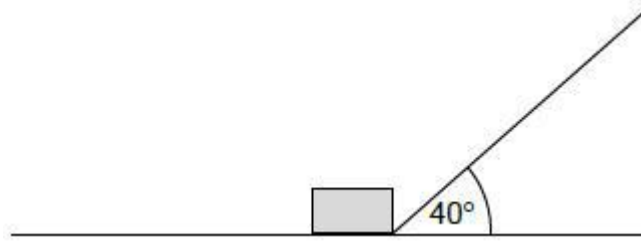
Q8.

In this question use $g = 9.8 \text{ m s}^{-2}$, giving your final answers to an appropriate degree of accuracy.

The diagram shows a box, of mass 8.0 kg, being pulled by a string so that the box moves at a constant speed along a rough horizontal wooden board.

The string is at an angle of 40° to the horizontal.

The tension in the string is 50 newtons.



The coefficient of friction between the box and the board is μ

Model the box as a particle.

- (a) Show that $\mu = 0.83$

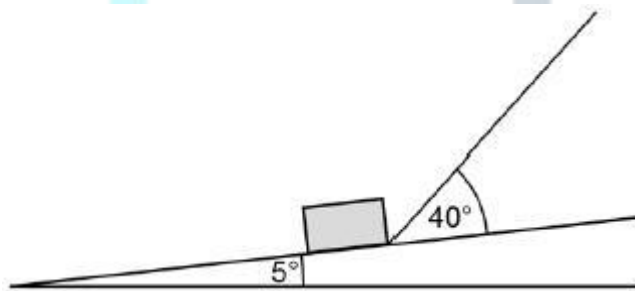
(4)

- (b) One end of the board is lifted up so that the board is now inclined at an angle of 5° to the horizontal.

The box is pulled up the inclined board.

The string remains at an angle of 40° to the board.

The tension in the string is increased so that the box accelerates up the board at 3 m s^{-2}



- (i) Draw a diagram to show the forces acting on the box as it moves.

(1)

- (ii) Find the tension in the string as the box accelerates up the slope at 3 m s^{-2} .

(7)

(Total 12 marks)

Q9.

A block of mass 30 kg is dragged across a rough horizontal surface by a rope that is at an angle of 20° to the horizontal. The coefficient of friction between the block and the surface is 0.4 .

- (a) The tension in the rope is 150 newtons .

- (i) Draw a diagram to show the forces acting on the block as it moves.

(2)

- (ii) Show that the magnitude of the normal reaction force on the block is 243 newtons , correct to three significant figures.

- (3)
- (iii) Find the magnitude of the friction force acting on the block.
- (2)
- (iv) Find the acceleration of the block.
- (4)
- (b) When the block is moving, the tension is reduced so that the block moves at a constant speed, with the angle between the rope and the horizontal unchanged. Find the tension in the rope when the block is moving at this constant speed.
- (5)
- (c) If the block were made to move at a greater **constant** speed, again with the angle between the rope and the horizontal unchanged, how would the tension in this case compare to the tension found in part (b)?
- (1)
- (Total 17 marks)

Q10.

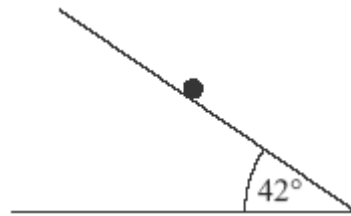
A particle of mass 3 kg is on a smooth slope inclined at 60° to the horizontal. The particle is held at rest by a force of T newtons parallel to the slope, as shown in the diagram.



- (a) Draw a diagram to show all the forces acting on the particle.
- (1)
- (b) Show that the magnitude of the normal reaction acting on the particle is 14.7 newtons.
- (2)
- (c) Find T .
- (2)
- (Total 5 marks)

Q11.

A particle, of mass m kg, is at rest on a rough plane which is inclined at an angle of 42° to the horizontal, as shown in the diagram.

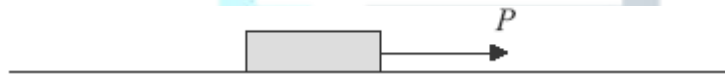


The friction force acting on the particle has magnitude 30 newtons.

- (a) Draw and label a diagram to show all the forces acting on the particle. (1)
 - (b) Find m . (3)
 - (c) Find the magnitude of the normal reaction force acting on the particle. (2)
 - (d) Given that the particle is on the point of sliding down the plane, find the coefficient of friction between the particle and the plane. (3)
- (Total 9 marks)**

Q12.

A block, of mass 10 kg, is at rest on a rough horizontal surface, when a horizontal force, of magnitude P newtons, is applied to the block, as shown in the diagram.



The coefficient of friction between the block and the surface is 0.5.

- (a) Draw and label a diagram to show all the forces acting on the block. (1)
 - (b)
 - (i) Calculate the magnitude of the normal reaction force acting on the block. (1)
 - (ii) Find the maximum possible magnitude of the friction force between the block and the surface. (1)
 - (iii) Given that $P = 30$, state the magnitude of the friction force acting on the block. (1)
 - (c) Given that $P = 80$, find the acceleration of the block. (3)
- (Total 7 marks)**

Q13.

A motorcycle and rider, of total mass 300 kg, are accelerating in a straight line along a horizontal road at 2.2 m s^{-2} .

- (a) Show that the magnitude of the resultant force acting on the motorcycle is 660 N.

(1)

- (b) A forward driving force of P newtons together with a resistance force of magnitude 400 newtons act on the motorcycle. Find P .

(2)

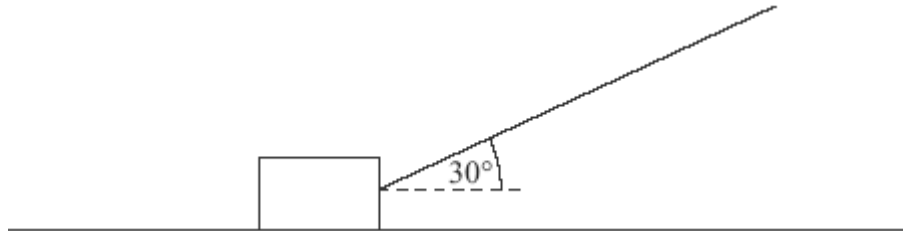
- (c) Find the time that it would take for the speed of the motorcycle to increase from 12 m s^{-1} to 23 m s^{-1} .

(3)

(Total 6 marks)

Q14.

The diagram shows a block, of mass 20 kg, being pulled along a rough horizontal surface by a rope inclined at an angle of 30° to the horizontal.



The coefficient of friction between the block and the surface is μ . Model the block as a particle which slides on the surface.

- (a) If the tension in the rope is 60 newtons, the block moves at a constant speed.

- (i) Show that the magnitude of the normal reaction force acting on the block is 166 N.

(3)

- (ii) Find μ .

(4)

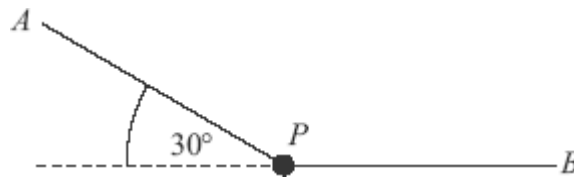
- (b) If the rope remains at the same angle and the block accelerates at 0.8 m s^{-2} , find the tension in the rope.

(5)

(Total 12 marks)

Q15.

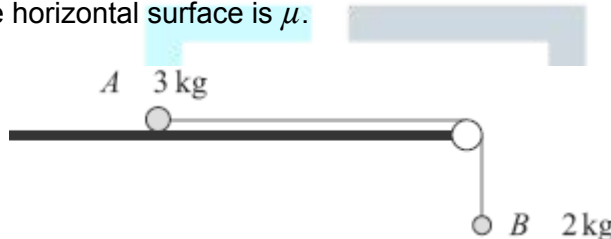
A particle, of mass 4 kg, is suspended in equilibrium by two light strings, AP and BP . The string AP makes an angle of 30° to the horizontal and the other string, BP , is horizontal, as shown in the diagram.



- (a) Draw and label a diagram to show the forces acting on the particle. (1)
- (b) Show that the tension in the string AP is 78.4 N . (3)
- (c) Find the tension in the horizontal string BP . (2)
- (Total 6 marks)

Q16.

Two particles, A and B , are connected by a light inextensible string, which passes over a smooth peg. Particle A is on a rough horizontal surface and has mass 3 kg . Particle B hangs freely, as shown in the diagram, and has mass 2 kg . The coefficient of friction between A and the horizontal surface is μ .



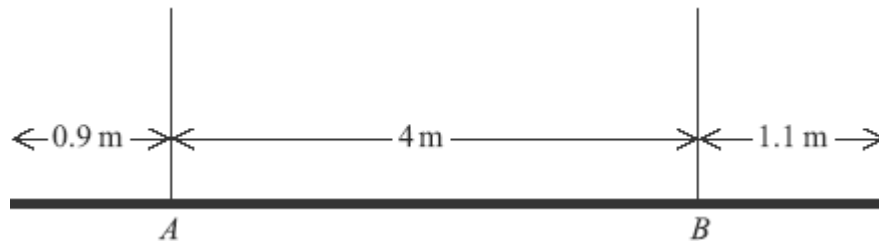
The particles are released from rest and move with a constant acceleration of magnitude 0.9 m s^{-2} .

- (a) Find the tension in the string. (3)
- (b) Draw and label a diagram to show the forces acting on particle A . (1)
- (c) Calculate the magnitude of the normal reaction force acting on A . (1)
- (d) Find the magnitude of the friction force that acts on A . (2)
- (e) Find μ . (2)
- (Total 9 marks)

Q17.

A uniform plank, of length 6 metres , has mass 40 kg . The plank is held in equilibrium in a

horizontal position by two vertical ropes attached to the plank at A and B , as shown in the diagram.



- (a) Draw a diagram to show the forces acting on the plank. (1)
 - (b) Show that the tension in the rope attached to the plank at B is $21g\text{ N}$. (3)
 - (c) Find the tension in the rope that is attached to the plank at A . (2)
 - (d) State where in your solution you have used the fact that the plank is uniform. (1)
- (Total 7 marks)**

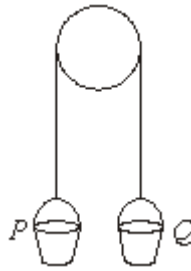
Q18.

A hot air balloon is at rest on the ground. When the balloon is released, it rises to a height of 320 metres in 80 seconds. The balloon moves under the action of its weight and a vertical lift force. Assume that the balloon has a constant acceleration during this motion.

- (a) Show that the acceleration of the balloon is 0.1 m s^{-2} . (3)
 - (b) Find the speed of the balloon when it reaches a height of 320 metres. (2)
 - (c) The mass of the balloon is 450 kg. Show that the magnitude of the vertical lift force is 4500 N, correct to two significant figures. (3)
 - (d) After a while, the vertical lift force is reduced so that the balloon rises at a constant speed. State the magnitude of the vertical lift force when this is the case. (1)
- (Total 9 marks)**

Q19.

A builder ties two identical buckets, P and Q , to the ends of a light inextensible rope. He hangs the rope over a smooth beam so that the buckets hang in equilibrium, as shown in the diagram.



The buckets are each of mass 0.6 kg.

- (a) (i) State the magnitude of the tension in the rope. (1)
- (ii) State the magnitude and direction of the force exerted on the beam by the rope. (2)
- (b) The bucket Q is held at rest while a stone, of mass 0.2 kg, is placed inside it. The system is then released from rest and, in the subsequent motion, bucket Q moves vertically downwards with the stone inside.
- (i) By forming an equation of motion for each bucket, show that the magnitude of the tension in the rope during the motion is 6.72 newtons, correct to three significant figures. (6)
- (ii) State the magnitude of the force exerted on the beam by the rope while the motion takes place. (1)
- (Total 10 marks)

Q20.

A crate is being pulled at constant speed across rough horizontal ground by a rope.

The crate is of weight 100 newtons and the frictional force between the crate and the ground is of magnitude 30 newtons.

The tension in the rope is of magnitude T newtons.

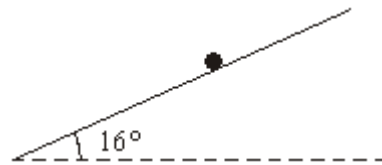


- (a) Draw and label a diagram to show all the forces acting on the crate. (1)
- (b) The coefficient of friction between the crate and the ground is 0.5. Show that the normal reaction force between the crate and the ground is 60 newtons. (2)

- (c) Explain why the horizontal component of the tension in the rope is 30 newtons. (2)
- (d) Find the value of T . (4)
- (e) Find the angle that the rope makes with the horizontal. (3)
- (Total 12 marks)**

Q21.

A stone rests in equilibrium on a rough plane inclined at an angle of 16° to the horizontal, as shown in the diagram. The mass of the stone is 0.5 kg.

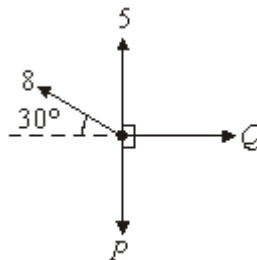


- (a) Draw a diagram to show the forces acting on the stone. (1)
- (b) Show that the magnitude of the frictional force acting on the stone is 1.35 newtons, correct to three significant figures. (3)
- (c) Find the magnitude of the normal reaction force between the stone and the plane. (2)
- (d) Hence find an inequality for the value of μ , the coefficient of friction between the stone and the plane. (2)
- (Total 8 marks)**

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Q22.

A particle is in equilibrium under the action of four horizontal forces of magnitudes 5 newtons, 8 newtons, P newtons and Q newtons, as shown in the diagram.



- (a) Show that $P = 9$. (3)

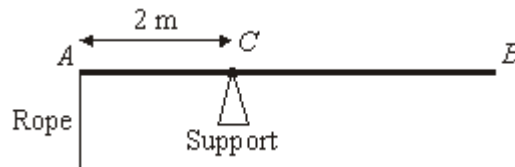
- (b) Find the value of Q .

(2)

(Total 5 marks)

Q23.

The diagram shows a uniform rod, AB , of mass 10 kg and length 5 metres. The rod is held in equilibrium in a horizontal position, by a support at C and a light vertical rope attached to A , where AC is 2 metres.



- (a) Draw and label a diagram to show the forces acting on the rod.

(1)

- (b) Show that the tension in the rope is 24.5 N.

(3)

- (c) A package of mass m kg is suspended from B . The tension in the rope has to be doubled to maintain equilibrium.

- (i) Find m .

(4)

- (ii) Find the magnitude of the force exerted on the rod by the support.

(3)

- (d) Explain how you have used the fact that the rod is uniform in your solution.

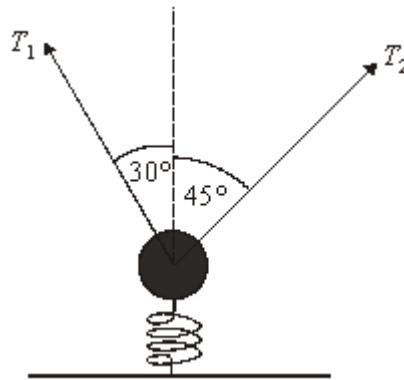
(1)

(Total 12 marks)

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Q24.

Two ropes are attached to a load of mass 500 kg. The ropes make angles of 30° and 45° to the vertical, as shown in the diagram. The tensions in these ropes are T_1 and T_2 newtons. The load is also supported by a vertical spring.



The system is in equilibrium and $T_1 = 200$.

- (a) Show that $T_2 = 141$, correct to three significant figures.

(3)

- (b) Find the force that the spring exerts on the load.

(4)

(Total 7 marks)

Q25.

A sphere of mass 200 grams is released from rest and allowed to fall vertically.

- (a) A student states that the acceleration of the sphere is 9.8 m s^{-2} while it is falling. What modelling assumption is this student making?

(1)

- (b) The student conducts an experiment and finds that the acceleration of the ball is in fact 8 m s^{-2} . He formulates a model for the motion that assumes a constant resistance force acts on the ball as it is falling.

- (i) Calculate the magnitude of this resistance force based on this assumption.

(3)

- (ii) Describe how the resistance force would vary in reality.

(1)

- (c) In a revised model the resistance force is assumed to be proportional to the speed of the sphere.

- (i) State the initial acceleration of the sphere.

(1)

- (ii) State what would happen to the acceleration of the sphere if it were able to fall for a long period of time.

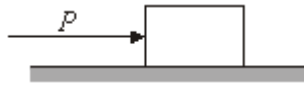
(1)

(Total 7 marks)

Q26.

A small block, of mass 2 kg, is on a rough horizontal surface. The coefficient of friction

between the block and the surface is $\frac{1}{7}$. A horizontal force of magnitude P newtons is applied to the block, as shown in the diagram.



- (a) The magnitude of the frictional force that acts between the block and the surface is F newtons. Show that the maximum possible value of F is 2.8.

(2)

- (b) The value of P increases gradually from 0 to 5.

- (i) When $P = 1$, state the value of F .

(1)

- (ii) When $P = 5$, the block moves with constant acceleration $a \text{ m s}^{-2}$. Find the value of a .

(4)

(Total 7 marks)

Q27.

A car moves in a straight line up a hill which is inclined at an angle of α to the horizontal, where $\sin \alpha = \frac{1}{14}$. The mass of the car is 1200 kg.

- (a) During the first part of its motion, the car moves with constant speed and is subject to a propulsive force of 1500 newtons and a resistance force of R newtons.

- (i) Draw a diagram showing the forces acting on the car.

(2)

- (ii) Show that $R = 660$.

(4)

- (iii) The car travels 60 metres in 4 seconds. Write down the speed of the car.

(1)

- (b) During the second part of its motion, the propulsive force ceases to act and the resistance force is still 660 newtons. The car continues to move up the hill until it comes to rest.

- (i) Find the magnitude of the retardation of the car in this stage.

(2)

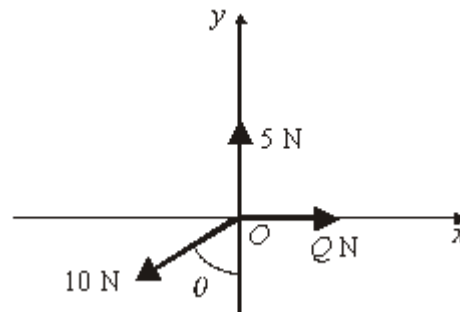
- (ii) Find the time between the propulsive force ceasing to act and the car coming to rest.

(2)

(Total 11 marks)

Q28.

A particle rests in equilibrium at a point O on a horizontal surface. It is acted on by three horizontal forces of magnitudes 5 N, 10 N and Q N. The directions of these forces, relative to the horizontal axes Ox and Oy , are shown in the diagram below.



- (a) Show that $\theta = 60^\circ$.

(2)

- (b) Find the value of Q .

(2)

(Total 4 marks)

Q29.

A children's slide is straight and inclined at 25° to the horizontal, as shown in the diagram.



Matthew, of mass 35 kg, goes down the slide at constant speed.

- (a) Draw a diagram to show the forces acting on Matthew.

(1)

- (b) Find the magnitude of the normal reaction force between Matthew and the slide.

(3)

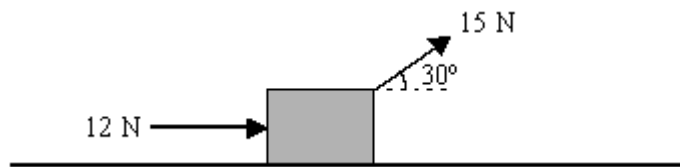
- (c) Find the coefficient of friction between Matthew and the slide.

(4)

(Total 8 marks)

Q30.

The diagram shows a small box resting on a rough horizontal surface.



The box is of weight W newtons. It is pushed with a horizontal force of 12 newtons, and pulled with a force of 15 newtons at an angle of 30° to the horizontal.

The box rests in limiting equilibrium.

- (a) Draw a diagram to show all the forces acting on the box. (2)
 - (b) Show that the frictional force acting on the box is approximately 25 newtons. (3)
 - (c) The coefficient of friction between the box and the surface is $\frac{1}{3}$.
Find the normal reaction force between the box and the surface. (2)
 - (d) Find the value of W . (3)
- (Total 10 marks)

Q31.

The diagram shows a car pulling a trailer in a straight line on a horizontal stretch of road.



The mass of the car is 1250 kg and the total resistance force acting on the car is 500 N.

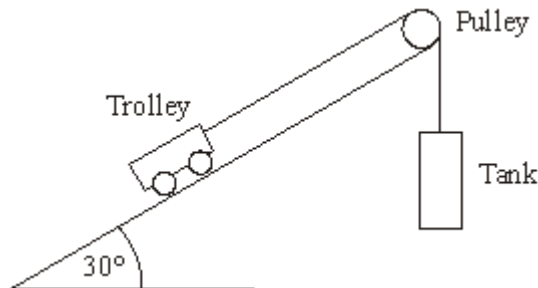
The mass of the trailer is 250 kg and the total resistance force acting on the trailer is 100 N.

- (a) During part of the journey, a constant braking force is applied to the car, causing the car and trailer to decelerate at a constant rate of 0.5 m s^{-2} .
 - (i) By considering the forces on the trailer, find the magnitude of the force in the towbar between the car and the trailer. (3)
 - (ii) Find the magnitude of the braking force applied to the car. (4)
- (b) Later in the journey, the car and trailer travel with constant speed.
State the magnitude of the tension in the towbar. (1)

(Total 8 marks)

Q32.

A trolley, of mass 200 kg, is on a slope inclined at 30° to the horizontal. It is attached to a tank by a light, inextensible rope that passes over a smooth, light pulley. The tank can be filled with water. The situation is shown in the diagram below.



Model the tank and trolley as particles and assume that there is no resistance to motion.

- (a) When the tank is filled with water, the total mass of the tank and the water is 150 kg.

Show that the acceleration of the tank and trolley is 1.4 m s^{-2} .

(7)

- (b) When the tank is empty, the trolley can be set in motion, so that it moves with a constant speed up the slope. Find the mass of the empty tank.

(4)

(Total 11 marks)

Q33.

A lift has mass 800 kg. A vertical cable is attached to the lift. Model the lift as a particle.

- (a) Show that the tension in the cable is 8000 N, when the lift is accelerating upwards at 0.2 m s^{-2} .

(3)

- (b) Find the tension in the cable when the lift is accelerating downwards at 0.2 m s^{-2} .

(2)

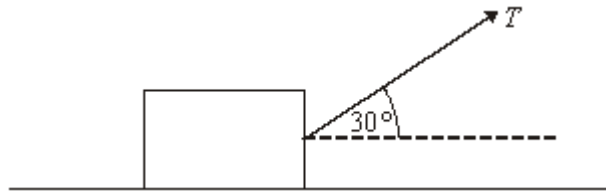
- (c) Write down the tension in the cable when the lift is travelling at a constant speed.

(1)

(Total 6 marks)

Q34.

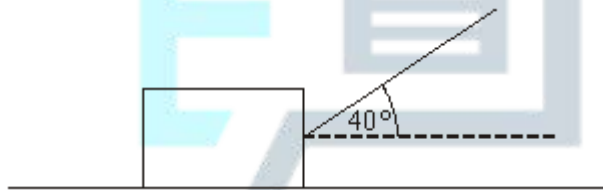
A heavy crate, of mass 200 kg, is pulled along a rough horizontal surface at a constant speed by a rope. The rope is at an angle of 30° to the horizontal. The tension in the rope is T newtons. The coefficient of friction between the crate and the surface is 0.6. Model the crate as a particle.



- (a) Draw a diagram to show the forces acting on the crate. (1)
- (b) Show that the magnitude of the normal reaction force on the crate is $(1960 - 0.5 T)$ newtons. (3)
- (c) Find T . (5)
- (Total 9 marks)

Q35.

A box, of mass 50 kg, is dragged along rough horizontal ground by a rope attached to the box. The rope makes an angle of 40° with the ground and the tension in the rope is T newtons. The coefficient of friction between the box and the ground is 0.6. The diagram shows the box and the rope.



Model the box as a particle.

- (a) Draw a diagram to show the forces acting on the box. (1)
- (b) Find, in terms of T , the magnitude of the normal reaction force acting on the box. (3)
- (c) Show that the magnitude of the friction force acting on the box is $294 - 0.6T \sin 40^\circ$ (2)
- (d) The box accelerates at 0.5 m s^{-2} . Find T . (4)
- (Total 10 marks)